PKCERT AI & Software Development Internship

Task 13 – Advanced Model Evaluation & Handling Imbalanced Data

Objective:
Develop practical skills in evaluating classification models using ROC-AUC curves, learning curves, and techniques for handling imbalanced datasets such as SMOTE and class weighting.

Total Marks: 100

Part A – Dataset Selection & Preparation (15 Marks)

1. Select a public classification dataset.

2. Describe the dataset, features, and target variable.

3. Preprocess the data and split it into training and testing sets.

Part B – Advanced Model Evaluation (35 Marks)

1. Build a classification model.

2. Generate and interpret the ROC Curve and ROC-AUC Score.

3. Generate a Learning Curve and discuss underfitting, overfitting, and generalization.

Part C – Handling Imbalanced Data (35 Marks)

1. Analyze class imbalance.

2. Apply SMOTE and retrain the model.

3. Apply Class Weighting and compare with the original model using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.

Part D – Comparative Analysis & Recommendation (15 Marks)

1. Compare all approaches and recommend the best-performing method with justification.

Assessment Area	Marks
Dataset Preparation	15
Advanced Model Evaluation	35
Handling Imbalanced Data	35
Comparative Analysis &	15

Recommendation	
Total	**100**